

DSAA3073: Theories in Data Science

Week 4 Open Investigation

When Does Model Selection Carry a Valid Certificate?

From empirical Rademacher penalties to robust structural risk minimization

Basic analysis. Selecting an emergency-department risk model

Use only the agnostic, Rademacher, symmetrization, and SRM tools in the Week 4 Learning Sheet

A hospital compares three nested classifier classes for predicting thirty-day emergency readmission. The audit sample contains $m = 5000$ i.i.d. patient records from the current deployment population. For each class, the training team reports the empirical risk of its ERM output and the empirical Rademacher complexity of the associated zero-one loss class:

$$\begin{aligned}\mathcal{H}_1 : \quad \hat{R}_S &= 0.180, \quad \hat{\mathfrak{R}}_S = 0.012, \quad w_1 = \frac{1}{2}; \\ \mathcal{H}_2 : \quad \hat{R}_S &= 0.130, \quad \hat{\mathfrak{R}}_S = 0.020, \quad w_2 = \frac{1}{3}; \\ \mathcal{H}_3 : \quad \hat{R}_S &= 0.100, \quad \hat{\mathfrak{R}}_S = 0.035, \quad w_3 = \frac{1}{6}.\end{aligned}$$

The confidence target is $1 - \delta = 0.95$. For class k , use $\delta_k = w_k \delta$ and the data-dependent one-sided radius

$$r_k = 2\hat{\mathfrak{R}}_{S(\mathcal{G}_k)} + 3\sqrt{\frac{\ln\left(\frac{2}{\delta_k}\right)}{2m}}.$$

Basic Task 1. Construct the simultaneous class certificates

Define the loss classes $\mathcal{G}_k$, compute every δ_k , r_k , and certified upper bound $\hat{R}_S + r_k$. Verify that the confidence allocation is valid and select the class prescribed by structural risk minimization. Explain why empirical risk alone makes a different choice.

Basic analysis. Interpret the selected certificate

The guarantee is simultaneous but conditional on its statistical environment

Basic Task 2. State the deployment claim and its limits

Write the probability statement that justifies the selected bound. Identify what is random, what is conditioned on when empirical Rademacher complexity is evaluated, and why the class index may be chosen after observing the data. Then give the hospital one justified statement and two conclusions that the calculation does not support.

Thinking investigation. Formalize the certificate

Separate the statistical class, optimization rule, and deployment target

Write the investigation as

$$(D, S, \{(\mathcal{H}_k, \mathcal{G}_k, w_k)\}_{k \in K}, A, \mathcal{C}),$$

where D is the audit distribution, A minimizes a penalized empirical criterion, and $\mathcal{C}$ is the simultaneous risk certificate. The basic theorem uses i.i.d. sampling, bounded measurable losses, valid classwise Rademacher bounds, and strictly positive confidence weights summing to one.

Investigation 1. Modify the confidence allocation

Suppose provisional active weights are positive and have finite sum $c > 0$, not necessarily one. If one sets $\delta_k = w_k \delta$ before normalizing, require every resulting class-level parameter to satisfy $0 < \delta_k < 1$. Under this validity condition, derive the certified failure bound and decide precisely what $c > 1$ does and does not imply. Then give a robust procedure for arbitrary positive provisional weights: normalize first, and construct the class radii only afterwards. Explain what happens to a class assigned weight zero.

Thinking investigation. Allow approximate penalized optimization

The statistical theorem survives with an explicit algorithmic tolerance

Suppose two-sided class radii r_k hold simultaneously: $|R(h) - \hat{R}_{S(h)}| \leq r_k$ for every $h \in \mathcal{H}_k$. Instead of exact SRM, the implementation returns $(\hat{k}, \hat{h})$ satisfying

$$\hat{R}_{S(\hat{h})} + r_{\hat{k}} \leq \inf_{k, h \in \mathcal{H}_k} (\hat{R}_{S(h)} + r_k) + \eta.$$

Investigation 2. Derive the approximate-SRM oracle inequality

Decide whether exact minimization is essential. Starting from the simultaneous two-sided event, derive an analytic population-risk bound for $(\hat{k}, \hat{h})$ in terms of every comparator (k, h) and the optimization tolerance η . Identify exactly how many times each class radius enters.

Thinking investigation. Change the deployment distribution

Generalization under the audit law is not automatically transfer

Investigation 3. Deploy under a shifted patient population

Let D be the audit distribution and Q the future deployment distribution. Define

$$\Delta_{\mathcal{G}}(D, Q) = \sup_{g \in \mathcal{G}} |\mathbb{E}_{D[g]} - \mathbb{E}_{Q[g]}|.$$

Decide whether the original Rademacher certificate for D remains a certificate for Q . If a finite upper bound Δ on this discrepancy is available, derive the corrected analytic expression. If no restriction links D and Q , determine whether deployment risk is identifiable from the audit sample.

Synthesis. State the full contract for responsible model selection

Return to the hospital case. Summarize the roles of empirical fit, Rademacher complexity, confidence allocation, optimization accuracy, and distribution stability. State the minimum additional evidence required if the hospital changes its patient population.